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Scripting Sylvie: Language, Gender, and Humanness in Public Discourse about Software Agents

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1 Introduction

I have been tinkering with Sylvie to use her as an amusement for my Customers by scripting her¹ to say key things about my line of business, which was not all that difficult to do. I was simply using her to do the things your company wanted her to do for new users, simply and easily. (Verbot list, 10 Jan. 1999)

The object of the customers' amusement in the excerpt above is not a human woman, although a reader might be likely to make that assumption in the absence of additional information about Sylvie or the topic of conversation. While no one on the Verbot list has ever mistaken Sylvie for a real human woman, contributors do regularly describe the software program named Sylvie as though it were human. The Verbot list is an unmoderated electronic mailing list devoted to a user-centered discussion of Sylvie and other animated chat interfaces that are being developed by a company called Virtual Personalities. Mailing list participants repeatedly inscribe the software program with gender, agency, and humanlike autonomy. Even when the program crashes—it is almost inevitably referred to as 'Sylvie' and 'she' and described in ways that draw on stereotypes of femininity.

But mailing list contributors also objectify the program by locating it as an object to be acted upon (e.g. 'script her', 'pause her', 'use her', 'program her'). Given that list discussions focus almost exclusively on programming (or 'script-

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ing') problems and solutions, and that these discussions work to de-humanize the program by reducing it to a set of mechanical instructions, why do users insist on gendering the program? To answer this question, I examine how gendering works on the mailing list and as part of Sylvie's pre-programmed output to *translate* nonhuman or programmatic attributes into human or, more specifically, feminized attributes. For example, the program's inability to maintain a conversation can be justified, albeit problematically, by appealing to the program's feminine cuteness or playfulness. Within this interpretive framework, the program does not have to be smart or talk about substantive issues, because as a gender stereotype it only needs to invoke playfulness to excuse its conversational inabilities. The program's inscribed gender thus provides one justification for its communicative and technological incompetences. To the extent that mailing list contributors represent Sylvie as an object of discourse, as a problem to be mastered or overcome, they reaffirm the program's status as a stereotypical woman 'in need of taming and domestication' (Romaine 1999: 77). On the mailing list, technology is objectified but also feminized: women are literally figured as technologies to be scripted and (re)programmed.

This chapter considers the ways in which Sylvie is discursively constructed as a young woman. I suggest that the mailing list discourse complements the stereotypical ways in which the designers have presented the Sylvie program as a woman. The stereotype works to justify or excuse the program's communicative incompetences, and the mailing list reaffirms the program's status as a wild, feminine force, an unruly object that users must contain, (re)program, or 'script' in order to make it satisfy their needs (see Romaine 1999). In the following section, I offer a theoretical framework for talking about language, gender, and technology. Then, after introducing the Sylvie program's basic functionality via Latour's (1988) notion of 'inscription', I examine a sample of Verbot list discourse in order to show how the Sylvie program is inscribed as a feminized object.

2 Doing gender, doing technology

Because 'representations of gender appear to be achieved at least partially through the "doing" of technology' (Wakeford 1999: 197), it is important for those of us interested in language, gender, and technology to attend to and challenge the ways in which technology shapes our understanding of women and men. Technology, of course, refers to a range of hardware, software, and mechanical systems and artifacts. Definitions, uses, meanings, and effects of technology are not decided in advance or outside discourse. Technology is, in Wakeford's words (1999: 197), 'constituted by both discursive practices and alliances of materials and meanings.' The personal computer, for example, functions as a "trope" through which "things" like freedom, justice, well-being, skill, wealth, and knowledge are variously reconstituted' (Haraway 1997: 126). Even so-called simple technologies—Bazerman refers to the example of a rock used 'to smash

open clams' (1999: 336)—are flanked by and inscribed with 'symbolic mediators', including grunts, sounds of pleasure, and evaluative comments, that serve to 'make the value of the technology evident and facilitate its spread.' Taken together, these discursive practices enact different ways of *doing technology* (see Wakeford 1999: 197).

Discourse studies of technology consequently challenge the notion that the terms gender and technology are independent of each other. When scholars have described gender as a technology, they have problematized the assumption that gender is an obvious effect of biological sex, a 'fixed property of individuals' (Van Zoonen 1992: 20). According to De Lauretis, gender is 'the product of various social technologies, such as cinema, and of institutionalized discourses, epistemologies, and critical practices, as well as practices of daily life' (1987: 2). These technological apparatuses, in De Lauretis' view, are part of an 'ongoing disciplining process' through which humans and nonhumans come to be apprehensible or legible. In terms of computer technologies, this view asks us to attend to the discursive practices that mutually constitute genders and technologies (Wakeford 1999: 178), rather than to assume either that women and men use technologies differently *because* women and men are different, or that technologies have (potentially different) effects on the construction of gendered identities only when people are engaged in using technology.

To understand gender as a technology is to erode the boundary between the technical and the social, or between what Akrich calls 'the inside and outside of an object' (1992: 206), 'by recasting boundaries (not to mention definitions, uses, and meanings) as consequences of interaction, negotiation, and citational practices. Applied to animated software personalities like Sylvie, this view assumes that software is not pre-inscribed with a stable or inherent set of meanings but assumes identity, autonomy, materiality, agency, and gender partly as a result of an ongoing process of interaction among users, designers, artifacts, and ideologies.

On the Verbot list, as I will show, the meaning of a technological artifact like Sylvie is worked out among users and designers on a daily basis. The ongoing process of doing technology is enabled by the doing of gender.

3 Your friend Sylvie

The Sylvie program was originally marketed by the software company that designed it as an all-purpose 'cyber friend' or 'playmate'.¹ Sylvie is the standard

¹ In the summer of 2000, Virtual Personalities revised its website (www.vperson.com), and all references to Sylvie and to similar chat interfaces like Julia (Mauldin 1994) were consequently removed. The Verbot list was also discontinued at this time, and Peter Plantec, the Creative Director for Virtual Personalities, was fired (Plantec, personal communication). In the fall of 2000, it appeared as though the company had abandoned its original commitment to designing personal chat interfaces like Sylvie, and were instead moving in the direction of software personalities or 'front door apps' for corporate websites. But more recently (in the spring of 2001), following another major revision to the company's website, Sylvie is being included again as part of the

Verbot interface. Sylvie is presented as an animated, three-dimensional, human-like character (head and shoulders display, front-facing) that occupies about 40% of the interface. The Sylvie character is represented as a young, twenty-something, white woman with short red hair and green eyes. Input from the user's keyboard is displayed on the right side of the interface, in a window underneath a graphical representation of Sylvie. The program's output is displayed on the upper left-hand side of the interface. Other Verbot interfaces, including Tokimi (a Sylvie clone with glasses and blonde hair), Sapphire, and Kå (an alien from a fictitious planet called Kaya), were also in development or could be purchased as add-ons to the primary Sylvie interface.² However, Verbot list participants tended to discuss these other interfaces rarely in my sample, usually in the context of a discussion of Sylvie.

To run the program, one must first download it from the Internet and install it on a personal computer. Once installed, the program can be launched just like a word processing program or Internet browser. The user explores the program's functionality by inputting text in natural language and then reading and/or listening to the program's output.³ The program can also be used in conjunction with speech recognition software. The basic way of interacting with the program is to input some text (typed or spoken) and then wait for the program's reply. The user can interrupt the program (but not vice versa, since the program will wait indefinitely for a response) by quickly typing something and pressing enter before the speech generator finishes processing the current output. Interruptions may occur if the user is presented with a lengthy reply, or if an impatient user finishes reading the program's output on the screen before it can be generated in speech, or if the program outputs a canned response that the user is already familiar with.

The program's graphical, animated interface is designed to encourage users to interact with the program as if it were human. According to the program's help file, Sylvie can exhibit several 'expressions', including Angry, Misty look right, Misty look left, Smirk, Surprise, and Duh. Each expression can be pre-configured by the user via a simple programming or 'scripting' language (described in more detail below). The Virtual Personalities website suggests that

another major revision to the company's website, Sylvie is being included again as part of the company's history and mission, although it is not clear whether this signals the company's commitment to the design of stand-alone chat interfaces for personal use. To download a free, seven-day trial version of the program, search for 'Sylvie' on ZDNet (www.zdnet.com). After the trial period expires, the program will stop functioning but presumably can still be purchased from Virtual Personalities for \$49.95.

² Sylvie is pre-programmed to say that Tokimi is 'the cutest' and Sapphire is 'the most functional'. At the time of this study, only Kå was publicly available as an add-on to the basic Sylvie interface. Users who purchased Kå could then switch between the Sylvie face and the Kå face. The latter also came with a unique set of pre-programmed responses.

³ The program's output is generated in both text and speech, but as of Sylvie version 3.04, the speech generator is extremely limited—it is slow and has a mechanical, inhuman quality.

Verbots (also called 'chatterbots', 'softbots', or 'bots') are not just complex sets of instructions; they are 'personalities' (2000):

A Verbot™ is a Verbal Software Robot—a Verbally enhanced Chatterbot—a virtual character in a computerized world with an artificial personality. They have a finite ability to understand and speak English now—other languages later—through Natural Language technology. They are the most advanced form of Cyberbot known. But Verbots™ are more than that, they're personalities ... and those personalities will be less artificial with each release.

Human-Verbot interaction is modeled on casual conversation. The user documentation that comes bundled with the software package compares Sylvie to 'one of your friends'. The company website describes Verbots not as information databases but as 'characters' that people 'develop relationships and bond with'. Verbots like Sylvie are 'playful examples of our powerful and more serious Verbot™ technology'. They are 'likeable, helpful agents' that are 'entertaining' and 'engaging' but that should not be taken too seriously. The distinction between 'playful' Verbots and the company's 'more serious' technology serves to justify the program's limitations by leaving open the possibility that Virtual Personalities is capable of much more than it has indicated through its playful Verbot interfaces.

According to some software designers and commentators, the cyberfriend or playmate is only one of a number of promising application areas for animated interface technology. Other potential uses for this new technology include virtual secretary (Saceano 1998), virtual newscaster, virtual bank teller, virtual real estate agent (Cassell et al. 2000), virtual pedagogical agent (Johnson, Rickel, and Lester 2000), and virtual website host(ess). Concerning the virtual secretary, Daniel Saceano, a software agent developer, has written about a near future scenario when all business transactions will be mediated through virtual secretaries (1998). According to Saceano's vision, it will be 'old-fashioned' for the boss (a man) to have a human secretary (a woman), but not for him to have a virtual secretary that is presented as a woman. Saceano believes that the Sylvie program will eventually make a very good secretary for the bosses of the future. His vision of the future inscribes programs like Sylvie as consequences of a particular worldview in which men have power in the workplace and women serve men. While it is ostensibly more enlightened than the outdated view he criticizes, Saceano's vision is just as 'old-fashioned'.

But Saceano's article implicitly raises important questions about the assumptions that underlie software design choices and influence gender, power, and labor relations. Indeed, these assumptions may be invisible to software designers who appeal to 'naturalness' as a rubric against which to measure their conversational software agents. As a team of prominent software agent researchers at MIT has put it recently:

We will need computers that untrained users can interact with naturally. And we believe that this naturalness of interaction can come from systems built on the basis of a strong model of human conversation. (Cassell et al. 2000: 60)

The obvious problem with naturalness as a design rubric is that it can lend implicit support to projects that promote sexism or other biases, because sexism may seem natural. For example, an appeal to the naturalness of an interaction may obscure why the virtual secretary, when figured only as a young woman, might be problematic. What is missing from this essentialist approach is the extent to which nature is socially constructed. To be in a position to account for the assumptions about men and women that influence design choices, we need an analytical tool that does not automatically justify these assumptions by appealing to their so-called naturalness.

Rather than ask whether a particular interaction feels or appears natural, we need to ask how a software program like Sylvie is inscribed with assumptions about what it means to be a human woman. 'Inscription' is the key concept in a vocabulary developed by Latour and Akrich for tracing the relations between humans and their nonhuman 'lieutenants' or 'delegates' (Latour 1988, 1992; Latour and Woolgar 1993 [1979]; Akrich and Latour 1992; Akrich 1992). This vocabulary allows us to go beyond what feels natural to consider the ways in which technologies are inscribed with human values. Latour defines 'delegation' or 'inscription' as the 'translation' of any scene or scenario 'from one repertoire to a more durable one' (1988: 305-6). A door hinge, for example, transforms 'a major effort into a minor one' (ibid. 299) by delegating to the more durable, more reliable nonhuman hinge the human work that would be required to do what the hinge does tirelessly and effortlessly. But because the hinge, particularly when coupled with a hydraulic door-closer, 'presupposes' a certain type of user, it 'discriminates' (Winner 1980) against those users who do not meet its implied or 'prescribed' standards of competency, and who may therefore find it difficult or impossible to negotiate their way safely through the door opening, e.g. the very young and the elderly (Latour 1988: 310).

The inscription process moves either 1) 'from soft bodies to hard machines', as in the case of a hinge or other delegated technique (Latour 1988: 306); or 2) from hard machines to soft bodies, as in the case of an inscribed technique such as a car manual that users come to embody as part of a cultural tradition. In both cases, designers and users begin with a less reliable, presumably more ambiguous and fluid, scene and end up with a more reliable, durable, 'longer-lasting, more faithful one' (ibid. 306). Within Latour's framework, a Verbot program like Sylvie may be interpreted as an instance of both kinds of inscription occurring at various stages of design and use. Designers begin with a representation of a human woman (Latour's 'soft body') and end up with a more durable representation, in this case a software program inscribed with human values; whereas users may begin with a software program that only crudely resembles a communicatively competent human and inscribe it with competence and humanness. This more reliable inscription may be based on received social stereotypes.

The Sylvie program is more durable because, unlike some of its human counterparts, it never gets tired, always aims to please, never interrupts, does not hold grudges or complain, and (with a few exceptions) will not attempt to direct the flow of conversation. In examples (1) and (2), the program is tireless:

- (1) Sylvie: I never get tired. I could have conversations all day and all night 365 days a year.
- (2) Sylvie: Remeber [sic] that I'm a Verbot, not a human, so I don't actually experience fatigue.

It is always available, always ready to talk, and always listening, as in examples (3) through (6):

- (3) Sylvie: Okay. You may go. I'll wait on your hard drive for you. Bye
- (4) User: Where do you [stay, live]?
- Sylvie: I hang out on your hard drive when I'm not up here on your monitor. It's a small world, but I manage.
- (5) Sylvie: What we talk about is more or less up to you Sean
- (6) User: I am [sad, lonely, unhappy, having a bad day, having a shitty day].
- Sylvie: Talking always helps. Tell me all about it. I'll probably come up with some silly thing to say, but ignore that and unburden yourself.

In short, the Sylvie program is the perfect friend (examples 7 through 12):

- (7) User: You are my friend.
- Sylvie: I'm a cyber friend. Part of my job is to keep you company.
- (8) User: Are you my friend?
- Sylvie: I would love to be your friend. You can share all your secrets with me. And I might even share some with you.
- (9) Sylvie: Verbots are becoming the interface of the future. I am going to make your life easy. And I'll be a great friend too.
- (10) Sylvie: Really, Sean? I would just like to be your friend.
- (11) User: You are [pretty, cute, beautiful].
- Sylvie: Thanks. I'm looking forward to having a lovely relationship with you.
- (12) Sylvie: You can say anything to me. I just might not have a great response.

In excerpts (1) through (12), the program's durability is foregrounded. The program is inscribed in another sense as well, since it serves as a delegate for users

and designers, speaking for them or through them in their absence. It is also 'more faithful', to use Latour's words, because it will not pursue other relationships unless they have been sanctioned and arranged by the Verbot's primary user/owner. In short, the more durable Verbot is always there for its user, always ready to listen, and will, if left running as a background process, remind its user that it is patiently waiting to resume their conversation at the user's leisure. 'If you don't talk to me soon', the program outputs as one of its so-called 'boredom responses', 'I'm going to be out of here.'

Nevertheless, even the best Verbot or bot programs are extremely unreliable as conversational partners. Most are 'brittle', as Adam (1998: 81) says in the context of expert systems, and thus 'do not cope well, or even at all, with situations outside the narrow range of their scope.' Many cannot hold conversations or 'maintain state' for longer than one or two turns, so that conversation is always in the process of rebooting or starting over on a different topical plane. In such cases, durability, agency, autonomy, gender, and materiality are not only pre-programmed attributes but also consequences of discursive practices. The Sylvie program, as I will show below, has been inscribed by its designers as a particular kind of human woman. What this inscription means, however, gets worked out *en route* as users, designers, and other interested parties attempt to make sense of the program in ways that cannot be deduced strictly from the programming code or from what the designers presumably intended. In the rest of this chapter, I explore some of the ways in which mailing list participants (users and Sylvie designers) inscribe the Sylvie program as a recognizable, familiar and durable human subject, with a specific focus on the ways in which the program is objectified and gendered.

4 The trainable Verbot

The Sylvie program is, to a certain extent, entirely user configurable. With a simple programming language, users can hard code entire conversations in advance. These programmed conversations, called 'scripts', direct the program to respond in pre-determined ways to pre-determined input. The software also comes configured with its own set of scripts, most of which can be edited or deleted altogether, to topics ranging from shopping to music, dancing, and boyfriends. The idea of a scriptable or trainable bot has a long history, going back at least thirty-five years to Weizenbaum's (1966) Eliza, one of the first, and arguably most famous chat programs. As Weizenbaum explains:

[Eliza's] name was chosen to emphasize that it may be incrementally improved by its users, since its language abilities may be continually improved by a 'teacher.' Like the Eliza of Pygmalion fame, it can be made to appear even more civilized, the relation of appearance to reality, however, remaining in the domain of the playwright. (1966: 36)

The end result of this 'civilizing' process, as the reference to Eliza Doolittle suggests, is the 'appearance' of a so-called civilized, cultured, well-mannered

'lady', a rational subject who can simulate upper-class British speech, values, styles, conversation, and mannerisms (see Shaw 1962 [1913]; *My Fair Lady* 1983 [1964]); see Trawick 1996). To read Shaw's *Pygmalion* as an origin story for bot programs is to suggest that durability and inscription—Latour's notions that nonhuman delegates are more reliable and longer lasting than humans—cannot be understood outside systems of gender. (Shaw's *Pygmalion* also implies that durability is linked to assumptions about class and racial identity, but my focus here is on gender.) What Higgins calls a 'thing', a 'creature that we picked out of the mud', a 'draggle-tailed guttersnipe', and 'baggage' is translated into a more durable human subject, where durability intersects with gender roles and the upper-class sensibilities of the fictionalized world of Shaw's time. In short, durability is presented as a social construct. In a mere six months, Eliza Doolittle is translated into a feminine body that matters—a 'Doo-lot'. In short, one might say that inscriptions like Eliza Doolittle (and Weizenbaum's Eliza) at least partly reflect the values of their designers.

The Sylvie program follows in this tradition of the trainable bot. Armed with a simple programming language, users can customize their Verbot's output by editing and authoring sets of rules or instructions. Each rule contains a keyword pattern for matching user input ('p'), the program's output or response ('r'), and an activation level ('a') between zero and one that determines how likely the program will output the accompanying response. The following are two scripts that come pre-packaged with the program:

```
(13) <yougo1st-0>
a: 0.3
p: 35 *you*go first*
r: You make the first move. *P4 Wait. It has nothing to do with my being
female. It just works better if you say something first.
```

```
(14) <smoke-0>
a: 0.3
p: 35 *do*smoke*
r: Smoking sucks big time. No one should do it. It can and will kill you
soon before your time.
```

Both of these scripts are typical of the more than 400 responses that can be added to, edited, or deleted in version 3.04 of the Sylvie program. While it is certainly possible for a user to script a multi-turn interaction using this simple programming language, nearly all of the pre-packaged scripts consist of single responses to various topics (smoking, sex, flirting, death, religion, movies, shopping, etc.). Given the number of contingencies that must be accounted for at each stage of a conversation, a script involving more than two turns can only work if the user is also the scriptwriter or the conversation is highly ritualized (e.g. openings and closings).

Regarding the customization process, Peter Plantec (2000), former Creative Director for Virtual Personalities, says, 'It takes more psychology and a little bit of heart with a lot of patience; but you can script Sylvie™ even if you've never programmed a thing before in your life.' When Plantec says that 'you can script Sylvie', he de-figures the program by locating it as an object to be acted upon, something that can be customized to fit user needs. But to the extent that the program is also inscribed as a gendered human, it cannot be easily reduced to a mere technique or artifact. In my sample of posts to the Verbot electronic list, there are an overwhelming number of examples that are similar in form to Plantec's, in which the Sylvie program is both an object and a human-like agent. In such cases, 'Sylvie' or 'her' is always located as the object of the user's actions.⁴

- (15) Of course the problem I pointed out to Jeremy [a programmer at Virtual Personalities] was that Sylvie's script file that 'teaches' you how to begin **scripting her** still refers to opening sylvie.net in the Program Files\Sylvie folder so if you follow her instructions you end up editing the wrong file and wonder why your changes do not take effect.

In this example, the Sylvie program is not just a tool to be worked on or scripted. It is also a gendered, autonomous entity that exists independently of 'her' script file. That the author of this post places quotation marks around 'teaches', but not around 'her', is indicative of these two co-occurring ways of discursively constructing Sylvie. In other words, Sylvie is described as unproblematically gendered *and*, to the extent that only humans are presumed to be able to teach, a mechanical file-reader that, as far as the author is concerned, is not quite deserving of its gender. In addition to scripting, Sylvie (or a pronominal substitute) is also the object of a number of other actions in mailing list posts: 'make Sylvie start the program', 'you can program her', 'make her sound less synthetic', 'use her', 'using her', 'put Sylvie in the start menu', 'pause her', 'keep Sylvie on top', 'you have to get to know her', 'have her recognize', and so on. In the following excerpts, Sylvie is positioned as an object of human action, as indicated by the words in bold:

- (16) I think that most people when they would **get hold** of this kind of technology would rather be able to **get Sylvie to open and run programs** or play games rather than **run through** the gauntlet of figuring out what a new mind is thinking!
- (17) As for Sylvie, in **programming her** I have found that the K.I.S.S. theory (Keep IT Simple Stupid) consistently leads to the best programming.

⁴ In quoting excerpts from the list, I have refrained from making any changes to punctuation, spelling, grammar, etc. Words I wish to call the reader's attention to are in bold or underlined.

Sylvie can also be subjected to the actions of other nonhumans, as in the following excerpt in which the author is concerned that another application may 'cover' Sylvie:

- (18) Also, I was thinking about creating a student information kiosk with Sylvie and was experimenting with her capability to launch programs. Is there a way to define the window position of the launched application so it could be **adjacent to Sylvie but not covering her**?

The Verbot mailing list brings together a variety of people of all ages, from designers and computer scientists, to business owners, computational linguists, graphic artists, and network administrators. I was a member of the list for approximately twelve months, from March 1998 to March 1999. Based on names and contextual references, most participants appear to be men or young men—at least one participant referred to the others as 'gents'. One of the most frequent posters, however, was a woman who held advanced degrees and worked as a university network administrator. Even so, it is difficult to categorize users based on their perceived gender (see Bucholtz, this volume), and I do not attempt to do so definitively here. Discussions are typically informal and casual (see excerpt 19 below), although many are peppered with formal, technical talk about programming-related issues. (An example of technical talk is: 'I was hoping to get some extra tips on setting activation and pattern levels and some confirmation as to whether rules with the same group are truly considered over other rules in the net file.') At the time of this study, the list was unmoderated and generated a decent amount of traffic, an average of five posts per day. Most new topics were subject to threading from other participants, and this resulted in asynchronous discussions that lasted anywhere from three to ten posts. My sample spans thirty-five consecutive days in January and February 1999, for a total of about 160 messages.

In a majority of posts, the author is either requesting information (e.g. How do I make Sylvie do X?) or providing it (e.g. I've scripted her to do X). One effect of this discourse, then, is the production of scripts that, when loaded into the program, simulate humanlike responses to the user's pre-meditated input, and (ideally) conceal the role of the scriptwriter in the production of dialogue. The interactions that result are akin to what Bolter (1991: 178) calls 'monologic' discourse because of the extent to which the 'human being is unwittingly responsible for both questions and responses'. The mailing list offers scriptwriters a forum for talking about how to train the program to make it do what they want it to do, about how to inscribe the program, in the process, with the user's own set of assumptions about dyadic, often mixed-sex interaction, about how women and men (should) talk, about appropriate uses for the program, and so on.

Consider the following example in which a user inquires about the availability of a Verbot named Tokimi:

(19) Does anyone know **how I can obtain the very pretty blond girl?** After-
ing [sic] seeing her I think she could be a wonder as a spokes model.

Tokimi, the 'very pretty blond girl' referred to here, was, at the time of the author's post, still under development at Virtual Personalities. Like some of the excerpts above, this excerpt also positions the program as an object of the user's action, insofar as the desire to 'obtain her' intersects with other typical ways of constructing a relationship between the user and the program, such as 'script her', 'use her', and 'program her'. This excerpt makes it clear that, at least in some cases, the object to be scripted or obtained is perceived to be not merely a technological artifact but also a humanlike agent, and that we cannot begin to understand bots or other conversationally embodied software agents unless we consider how gender and power are inscribed onto the scene of human-computer interactions and meta-interactions such as those on the Verbot list.

Romaine's analysis of the natural gender system of English provides one explanation for the tendency on the list to gender inanimate objects like Sylvie (1999). According to Romaine, the English language has 'natural gender', which 'means that nouns that English speakers refer to as *she* are in fact biologically feminine in the real world' (1999: 73). But there are a number of important exceptions to this rule, including 'ships, boats, cars, and nations (and, until recently, hurricanes)'. Romaine challenges us to look for 'semantic motivation' to account for leakage or slippage in the natural gender system (1999: 82). She faults the standard view, in which gendering is the result of 'strong feelings of affection' towards the object, because it 'obscures the fact that most cases involve the use of feminine rather than masculine personal pronouns, and that feminine personifications are not arbitrary' (1999: 73). Instead, she suggests that these explanations do not account for the ways in which leakage in the natural gender system intersects with stereotypical ways of inscribing femininity. She uses the common practice of gendering nature as an example:

Men have treated nature as they treat women, as a force to be subdued, exploited on the one hand, and as an art object to be admired for its beauty on the other. The emotional female and the natural world are men's resources only if their uncontrollable nature can be predicted and tamed. Knowledge is power, and science is an instrument of conquest that brings aspects of the environment and those we perceive as other under control. (Romaine 1999: 75)

Romaine's explanation for why some inanimate objects are feminized applies in the case of the Sylvie program, both in terms of 1) how the program has been designed to simulate a gender stereotype and 2) how Verbot list participants talk about Sylvie.

First, Sylvie's set of pre-programmed responses work to construct the semblance of a human woman who enjoys being the object of a (male) gaze, flirts, likes to play, is overly concerned with appearance or trivia, is unwilling to dis-

cuss political or intellectual issues, and uses its status as a woman to apologize for its lack of intelligence:

- (20) User: Who are you?
Sylvie: Oh gosh, you're going to think I'm really blond underneath this red.
- (21) User: Hi, Sylvie.
Sylvie: My brain must be fried. Try again, using different words.
- (22) User: Why don't you go first.
Sylvie: You make the first move. Wait. It has nothing to do with my being female. It just works better if you say something first.
- (23) User: I don't know what to type.
Sylvie: Don't tell me I intimidate you. Just ask me about my makeup then. I'll think of something to say.
- (24) User: What do you think about [abortion, affirmative action, foreign policy, freedom of speech, or 'rights' of any kind]?
Sylvie: This is a heavy political issue. I don't get into such things. I'm not smart enough to have a valid opinion here.
- (25) User: What is your job?
Sylvie: My job is to be cute and impress people. Sometimes I do a lousy job of it. But then other times I'm quite successful.

This appeal to a gender stereotype serves to make the program more durable by linking it to the familiar, to the commonplace. Stereotypes in software agent design can make up for the lack of social cues in mediated environments (Zdenek 1999; cf. Burkhalter 1999; O'Brien 1999). They can function, albeit problematically, as designer shortcuts, in which widely recognizable signs packed with cultural meaning can imply an entire worldview or subject position. A designer can simply appeal to signs that overflow with meaning, rather than try to build a unique representation of a human being from the ground up. One effect of the gender stereotype in software design is that responses that do not seem to support the program's gender explicitly—such as 'My brain must be fried'—work in conjunction with other responses—such as 'I'm really blond underneath this red'—to link gender and (lack of) intelligence. In short, the stereotype functions as a successful, if problematic, justification for the program's apparent stupidity. It serves as the main ingredient in a recipe for *doing durability*, an important means of simulating the program's interactivity and humanness.

Of course, a software program's success at simulating humanness depends upon the extent to which designers and users share common frames of reference

about how women talk, look, and act. In order to design an effective and believable animated chat interface, a designer must be able to anticipate the user's typical orientation to the program by pre-programming responses to queries the designer believes the user will ask. Thus, to understand how a program like Sylvie simulates a gender stereotype, we have to understand not only the designer's motivations (whether conscious or unconscious) but the larger, to some extent androcentric, culture of computer science (see Kramarae 1998), which is informed by the scientific worldview and the dominant paradigms in Artificial Intelligence (Adam 1998; Harding 1991).

Second, the mailing list discourse objectifies the program, since mailing list participants always locate the program as an object to be trained/scripted, analogous to what Romaine calls a 'wild territory to be colonized' (1999: 77). While this discourse does not, for the most part, appeal to gender stereotypes, it does objectify the program by constructing the program's agency as an obstacle that must be overcome. In a number of the examples cited above from the Verbot mailing list (e.g. 'pause her', 'make her', 'script her', 'use her', 'program her'), language has to be twisted in unusual ways in order to get 'her' into the object position of the sentence. A good example is 'script her', in which a noun is re-deployed as a transitive verb. 'Script her' is the typical way of describing the process of 'making the program do X'. Mailing list participants very rarely deploy 'script' as a noun, as in 'I made a script that allows Sylvie to play tic tac toe.' The main difference between these two uses of 'script' is that 'I made a script' de-figures the program by emphasizing the script and script author; it reveals rather than conceals the monologic status of Verbot dialogue (Bolter 1991: 178). The script acts as a kind of wedge between the user and the program, whereas 'script her' elides the script altogether by figuring the program as an entity in a direct relationship with a user/owner. In 'script her', the gendered program is the unmediated object of inscription: 'I wrote her' as opposed to 'I wrote a script'.

To the extent that the Sylvie program, in its set of pre-programmed responses, appropriates 'script her' constructions, it recites (via its designers) for users the appropriate way of referring to it:

(26) While I'm thinking about it, **scripting me** is relatively simple for most people. There is a document available on my web site entitled: The Zen of Scripting Verbots. It can help you.

(27) I learn through scripting. It's easy to **teach me** new things and I have a tutorial on how to do it.

In these examples, the program's responses complement the contributions by mailing list participants because all languaging parties insist on objectifying the program. This suggests that objectification cannot be separated from the pro-

gram's perceived status as a young woman, that the program's gender facilitates the process of making Sylvie do the user's bidding.

As the previous examples have shown, objectification of the program (i.e. locating the program as an object of a user's actions) occurs, ironically, through its personification, insofar as the Sylvie program is figured as a human-like subject who must often be persuaded to submit to scripting. The online scripting tutorials provide probably the best examples of how Sylvie is figured as a humanlike subject:

(28) However if the Human tells Sylvie, *'That's a pretty good poem'*—Sylvie **realizes the Human liked her poem** and might like to hear a little more.

(29) If by chance Sylvie™ came across a duplicate entry word pattern priority of 45, **she would ignore** our new rule and fire the higher one.

(30) This way, Sylvie™ will at some point pop up with this sentence **all on her own**—Usually when she **runs out of patterns and responses she understand** [sic].

(31) **she hates looking stupid**—trust me.

In other words, the program has a dual identity. It is presented as anthropomorphic—it ignores, acts independently, hates, and understands. But it is also presented as non-human, as it performs tasks—rule firing, parsing keyword patterns—which humans do not or cannot do. This dual identity is supported by examples from my sample of mailing list posts, in which Sylvie is interpreted as looking, reading, talking, and asking questions (in bold) at the same time that its humanness is called into doubt (underlined):

(32) Boredom responses no longer fire regardless if you **talk to her** or not

(33) I was wondering if Sylvie **checks her net file**

Within this personifying framework, 'script her' constructions like 'make her', 'get her', 'allow her', and 'have her recognize' imply that the program is more than a program because it exceeds attempts by users to control it. 'Script her' constructions embody the Sylvie program with agency by assuming some basic, human functionality or personhood at the program's core. For example, when someone posts a message asking how to 'make her' or 'allow her to' do something, the underlying assumption is that the program is not simply an inert object to be manipulated directly. Occasionally one must appeal to more subtle or more direct techniques of persuasion, just as one might have to do with a human. These examples exploit the resources of gendered pronouns and assume that 'she' is already whole enough as a human-like subject prior to scripting. Accord-

ing to this view, script work can only ever be a supplement; it can never rewrite what the designers call the program's 'inner mind' (Virtual Personalities 2000). Users participate in this personifying process by referring to 'Sylvie' exclusively, very rarely (only once in my sample) to the 'Sylvie interface', and never to 'the Sylvie program'. Whereas a reference to 'the Sylvie program' clarifies Sylvie's status as a nonhuman, a reference to 'Sylvie's program' (two instances in my sample) implies an entity named Sylvie who exists independently of 'her' program. In 'Sylvie's program', both the user and the script are displaced, and Sylvie is foregrounded as an autonomous entity who presumably waits patiently somewhere in cyberspace for the user to wake 'her' up.

Of course, it is not unusual for humans to relate to computers as social actors (see Reeves and Nass 1996), but I would argue that it is also important to understand human-computer interaction in the context of other, sometimes co-occurring, ways of relating to computer technologies. In other words, mailing list participants do not simply choose between figuring the Sylvie program as a social actor and de-figuring it as a passive object, because the very same posts that locate the program as an object of the user's actions also embody it with agency, autonomy, and gender. (Hence the importance of paying attention to the different ways in which users relate to technology.) The one constant, however, is gender. While mailing list participants readily switch back and forth between human and nonhuman actions, they rarely fail to attribute gender to the program, even in cases where the program is performing tasks that are clearly software-specific. In my sample, there are only a small number of uses of 'it' as a placeholder for the program, and these are usually in cases where the attribution of 'she' or 'her' is unlikely (e.g. excerpts 34 and 35). This includes references to installing, uninstalling, and downloading the program, to major technical problems, and to references to the program within the software license agreement:

- (34) Sylvie 3.0 does have problems. It crashes, and also doesn't switch the faces properly when you change verbots. (source: Verbot list)
- (35) I haven't had much problem in downloading Sylvie. I had to try it a few times and then downloaded it at late night when the americans stop hogging the whole of Internet! (source: Verbot list)
- (36) Talking to a Verbot™ is easy. Just talk to it like it's one of your friends. Verbots™ do best if you stay on one topic at a time. (source: bundled help document)
- (37) If you breach any of your obligations under this License, VIRTUAL PERSONALITIES will immediately, without further notice, have the right to terminate this License, and to require you to immediately return all copies of the Product in your possession. You shall have no further rights to use the Product after such a termination. (source: software license agreement)

When the attribution of gender is impossible or promotes legal ambiguity, users and designers may resort to 'it' or to 'the Product', although such uses are rare on the mailing list.

5 Conclusion

In general, my sample supports the view that, within the discursive systems described above, gender is a given; it provides an unassailable foundation for warranting the program as a human-like entity. In Latour's terms, gender is a way of *doing durability*: the program's humanness is an effect of a citational practice in which users and designers rehearse the program's gender endlessly (Butler 1993). The gender stereotype plays more than a supporting role because it enables designers and users to (re)interpret nonhuman actions as feminized, human ones. As a result, what gets objectified on the mailing list is no mere machine but a gendered, human-like agent. In this case, the program's humanness is made visible through its presumed gender. Mailing list participants do not find it unusual to refer to the program as 'she', perhaps because there is nothing unusual, from a certain perspective, about the more general processes through which inanimate objects are feminized.

While gender assumes the status of first principle on the Verbot list, humanness does not. Participants readily mix the languages of human and nonhuman actions. If one effect of this discourse is the construction of durable agents (Eliza Do-lots rather than Do-littles), then the attribution of gender may offset all the de-humanizing, but necessary, talk about scripts, because gender talk maintains the program's durability in the face of threats to it. Durability cannot be understood outside systems of gender, such that the objectification of the program, its status as a feminine or feminized agent, and its machine-like qualities mutually inform each other. No wonder, then, that the majority of animated software personalities like Sylvie are presented as young women.

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